

Endemic Nephropathies

Ramón García-Trabanino, Shuchi Anand, Richard J. Johnson, Magdalena Madero

Localized epidemics of chronic kidney disease (CKD) have been reported in various parts of the world and are known as *endemic nephropathies*. The word *endemic* is used in a lay rather than epidemiologic sense, to emphasize their regionally confined nature. By definition, these nephropathies are distinct from CKD of standard causes and tend to manifest insidiously. Although aging and the metabolic syndrome are also associated with reductions in estimated glomerular filtration rate (eGFR), endemic nephropathies are distinguished by earlier onset, absence of classic risk factors, and more rapid progression to kidney failure.

There have been remarkable examples in which the cause of endemic nephropathies have been identified. In the 1920s, an increased frequency of CKD in children and young adults in Queensland, Australia, was shown to be due to lead poisoning, such as from exposure to paints. Patients presented with either Fanconi syndrome from acute lead intoxication or CKD and hypertension associated with chronic intoxication. Likewise, an epidemic of CKD resulting from chronic interstitial nephritis and associated with Fanconi syndrome with hypophosphatemia and osteomalacia was observed in the 1950s and 1960s in the Toyama prefecture, Japan, and was shown to be caused by cadmium poisoning; a factory was dumping cadmium into the Jinzu River leading to contamination of the rice and soybeans. The disease was called *itai-itai* (“ouch-ouch”) by the locals due to the severe joint pains, and manifested over decades, especially among older women. For more information on nephropathies associated with heavy metals, see [Chapter 65](#). Finally, Balkan endemic nephropathy is another example in which the cause, the herb *Aristolochia clematis* contaminating wheat, has been identified. It is now considered to be an aristolochic acid–induced nephropathy, along with the Chinese herb nephropathy, a chronic tubulointerstitial disease observed in individuals taking these herbs for weight loss in the early 1980s. Diseases caused by *Aristolochia* (*Aristolochia*-associated nephropathy) are discussed in [Chapter 79](#).

MESOAMERICAN NEPHROPATHY

In 2002, physicians from the Hospital Rosales in San Salvador, El Salvador, reported a large number of patients presenting to the emergency room with kidney failure of undetermined etiology and a distinct epidemiologic pattern: most were Hispanic men between ages 20 and 60 performing manual work in the Pacific coast fields.¹ Subsequently, similar epidemic clusters were described along the Pacific coast of Guatemala, Nicaragua, and Costa Rica ([Fig. 66.1](#)). Sugarcane cutters have been the most studied, although the disease may be found among all types of farmers, miners, and construction and transportation workers, among others. Review of historical records suggest the phenomenon has been present since the 1970s, but diagnosis and registry has only been performed during recent decades ([Fig. 66.2](#)).² In some areas, such as in Chichigalpa, Nicaragua, the prevalence of CKD (defined as eGFR

<60 mL/min/1.73 m²) approaches 40% of men between ages 20 and 40 years.³ This contrasts with an overall prevalence of 8.3% for all adults and with less than 0.4% for men ages 20 to 40 years in the United States.

Pathogenesis

Although rare reports link Mesoamerican nephropathy with exposure to carbamate pesticides or methyl parathion, no confirmatory data exist.^{4,5} However, lag between exposure and disease, the variability in environmental fate and stability of agrochemicals, and changes in compounds over time, added to their seasonal use, make the contribution from agrochemicals difficult to evaluate. Heavy metals have been investigated as well, along with silica and nickel,^{6–8} but definitive evidence has not yet emerged.

One common risk factor is heat stress and recurrent dehydration, leading some to suggest this might be a type of heat stress nephropathy. Heat stress is known to cause acute kidney injury (AKI) from a variety of mechanisms, including heat stroke and rhabdomyolysis, but the concept of CKD occurring from heat exposure is new. Consistent with this possibility is the observation that Mesoamerican nephropathy is much more common in people who work at low altitudes, where it is warmer than at higher altitudes, despite similar working conditions and pesticide exposure.⁹ Experimentally, recurrent heat stress and dehydration have been shown to induce CKD in animals through hyperosmolarity-induced activation of the polyol-fructose and vasopressin pathways, and the kidney injury is amplified if animals are rehydrated with fructose-containing beverages as opposed to water. Kidney injury is also worse if body temperatures are higher in response to heat. Heat stress also may result in subclinical rhabdomyolysis, uric acid overproduction, and supersaturation of urine leading to intermittent uric acid crystalluria.¹⁰ Some infectious diseases, such as leptospirosis, have also been proposed as a causal or potentiating factor, although supporting evidence is limited.⁷

Pathology

The primary finding in patients with advanced CKD is chronic tubulointerstitial fibrosis with variable degrees of glomerulosclerosis ([Fig. 66.3](#)).¹¹ Whether the glomerular changes are secondary to chronic disease or represent a primary glomerular injury is unclear. Immune deposits are absent. A subset of patients may present with AKI and acute tubulointerstitial disease on biopsy.¹² Hyperuricemia, a common feature, has been found experimentally to cause glomerular hypertension, and vasopressin induces hyperfiltration in addition to the effects on tubular injury; however, uric acid crystals are not present in biopsies.

Clinical Manifestations

In early stages, some patients present with AKI or with muscle weakness related to hypokalemia or with aseptic dysuria, possibly related to uric acid crystalluria. Other asymptomatic patients are identified during

Mesoamerican Nephropathy Is Common in Hotter Areas of Central America

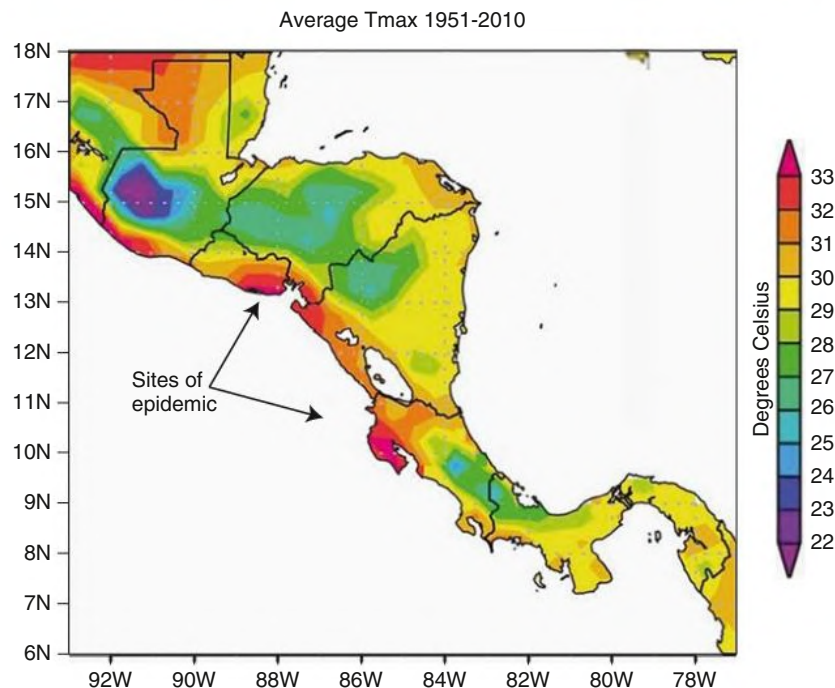


Fig. 66.1 Mesoamerican Nephropathy. Major areas where Mesoamerican nephropathy has been observed are along the Pacific Coast, especially involving the Guanacaste region of Costa Rica, the Bajo Lempa region of El Salvador, and the Chinandega region of Nicaragua. These are also the hottest areas in Central America. (From Glaser J, Lemery J, Rajagopalan B, et al. Climate change and the emergent epidemic of chronic kidney disease from heat stress in rural communities: the case for heat stress nephropathy. *Clin J Am Soc Nephrol*. 2016;11[8]:1472–1483.)

Epidemiology of Mesoamerican Nephropathy in Costa Rica

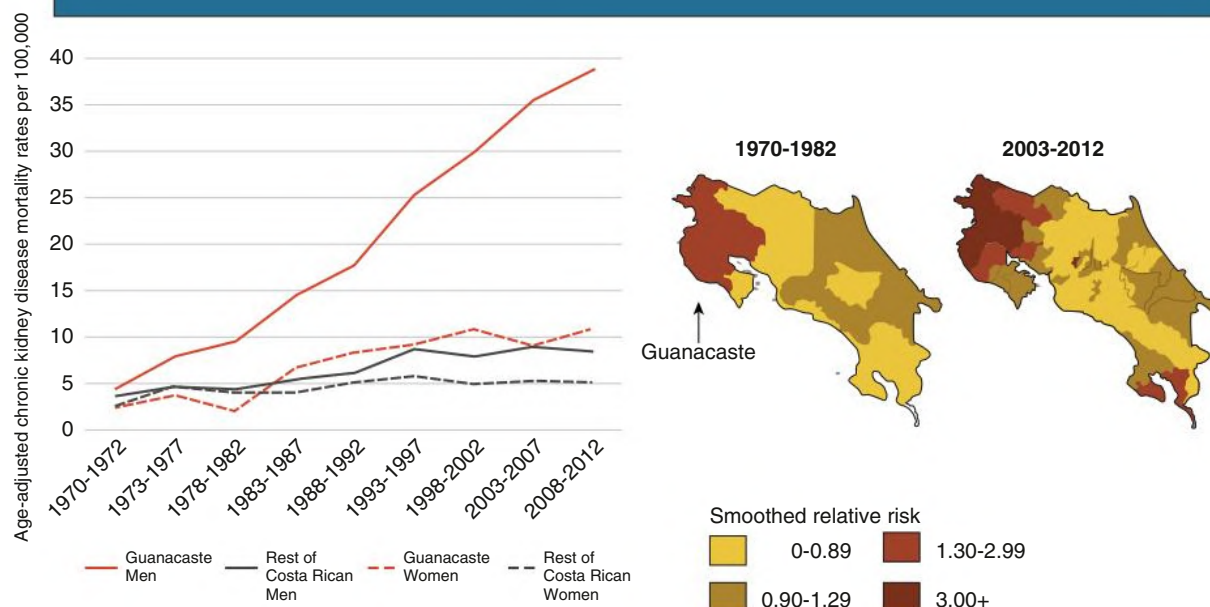


Fig. 66.2 Mortality of Chronic Kidney Disease in Guanacaste, Costa Rica. Mesoamerican nephropathy has been suspected since the 1970s in the Guanacaste region of Costa Rica. Shown is the increasing mortality of chronic kidney disease in the Guanacaste region of Costa Rica in men compared with the rest of Costa Rica. (From Wesseling C, van Wendel de Joode B, Crowe J, et al. Mesoamerican nephropathy: geographic distribution and time trends of chronic kidney disease mortality between 1970 and 2012 in Costa Rica. *Occup Environ Med*. 2015;72[10]:714–721.)

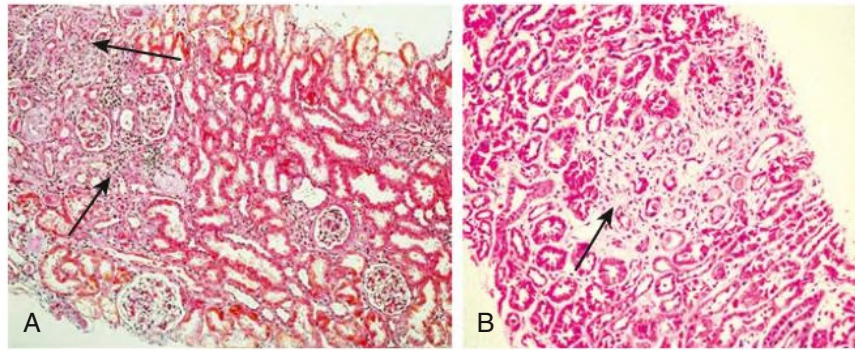


Fig. 66.3 Renal Pathology in Endemic Nephropathy. (A) Mesoamerican nephropathy is characterized by chronic tubulointerstitial fibrosis with localized inflammation (arrows). Other findings include glomerular ischemia and variable glomerulosclerosis. Renal vascular disease is usually mild. (B) Sri Lankan nephropathy also has a similar histologic appearance with the tubulointerstitial fibrosis with some pockets of inflammatory cells. (x40 magnification. B, H&E stain.) (A, Courtesy Dr. Annika Wernerson, Karolinska Institute Sweden. B, From Wijetunge S, Ratnatunga NV, Abeysekera TD, et al. Endemic chronic kidney disease of unknown etiology in Sri Lanka: correlation of pathology with clinical stages. *Indian J Nephrol.* 2015;25[5]:274–280.)

TABLE 66.1 Characteristics of Patients With Mesoamerican Nephropathy

No evidence for common causes of CKD	Absence of diabetes (blood glucose <125 mg/dL or 7 mmol/L) BP normal or only slightly high Minimal evidence for GN (no RBC casts, urine protein <2 g/d) No evidence for obstruction (ultrasound) No evidence for polycystic kidney disease (ultrasound)
Clinical presentation	Asymptomatic rise in serum creatinine Minimal albuminuria Frequent hyperuricemia Serum potassium level often lower than expected Biopsy usually shows chronic tubulointerstitial disease and variable glomerulosclerosis (often performed late in disease)

BP, Blood pressure; CKD, chronic kidney disease; GN, glomerulonephritis; RBC, red blood cell.

health screening through elevated serum creatinine (Table 66.1). Blood pressure is normal or only slightly elevated, and urine sediment shows low-grade (<1 g/d) or no proteinuria with occasional red cells and leukocytes. Serum sodium and potassium are often low and appear to be associated with increased urinary losses⁷; similarly, elevated magnesium and phosphate losses in the urine are also common, supporting a tubular defect.¹³ Serum uric acid is generally elevated, typically 2 mg/dL higher than expected for the eGFR.¹⁴

Diagnosis, Treatment, and Prognosis

Diagnosis is made on clinical grounds, based on living in an endemic area and presenting with CKD without clear cause (Fig. 66.4). The presence of hypokalemia and hyperuricemia and the absence of relevant proteinuria aids diagnosis. Kidney biopsies are rarely performed. Empiric treatment includes counseling on hydration and prevention of heat stress and avoidance of nonsteroidal antiinflammatory drugs. Given experimental data that rehydration with sugary beverages could worsen kidney injury, avoidance of soft drinks or sugary beverages as hydration drinks is recommended. Local nephrologists treat subjects empirically with low-dose angiotensin-converting enzyme

inhibitors (especially if they are no longer at risk for dehydration and hypokalemia is present) and with allopurinol and bicarbonate when hyperuricemia is present to reduce the formation of urine urate crystals and improve the aseptic dysuria. Dialysis is infrequently available because of social determinants; prognosis is guarded, and mortality is high.^{7,15}

SRI LANKAN NEPHROPATHY

Reports of an epidemic of CKD in the North Central Province of Sri Lanka began in the late 1980s. Regionally, this disease is simply labeled CKDu (chronic kidney disease of uncertain etiology). There are over 100,000 individuals affected, with an ongoing mortality of 5000 per year.¹⁶ Typically, affected individuals are men between the ages of 30 and 60 years who are working in the rice paddies under hot conditions. Women also may develop the disease. The clinical presentation is similar to that of Mesoamerican nephropathy, with low-grade to no proteinuria and an asymptomatic elevation in serum creatinine, although biopsy data indicate concomitant CKDu among people with prior diagnoses of diabetes and hypertension as well¹⁷ (see Table 66.1). Hyperuricemia is also common.¹⁰ Kidney biopsies early in the disease show acute inflammation and/or chronic tubulointerstitial fibrosis, whereas later biopsies also show increasing glomerulosclerosis (see Fig. 66.3).^{17,18} Many patients are poor and have little access to medical care, although the government has organized screening camps to detect disease earlier and is increasing dialysis capacity.^{16,19}

The primary hypothesis has been that Sri Lankan nephropathy is caused by an agrochemical toxin or heavy metal exposure such as cadmium or arsenic. Nevertheless, levels of heavy metals in the drinking water are within acceptable ranges,²⁰ and it has been difficult to incriminate any specific agrochemical such as glyphosate.¹⁶ For example, in an analysis of hair and nail samples of persons with biopsy-proven CKDu, concentrations of trace elements matched those measured in controls.⁵ A genome-wide analysis identified potential loci for susceptibility that map to tubular function.²¹

Preliminary data also suggest those who drink local well water are at higher risk for CKD, suggesting the presence of an unidentified toxin.²¹ Importantly, the risk for CKD was also increased in those highly exposed to the sun, those who worked more than 6 hours per day, and those who drank less than 3 L of liquids per day.²² Noteworthy,

Diagnosis and Management of Mesoamerican Nephropathy

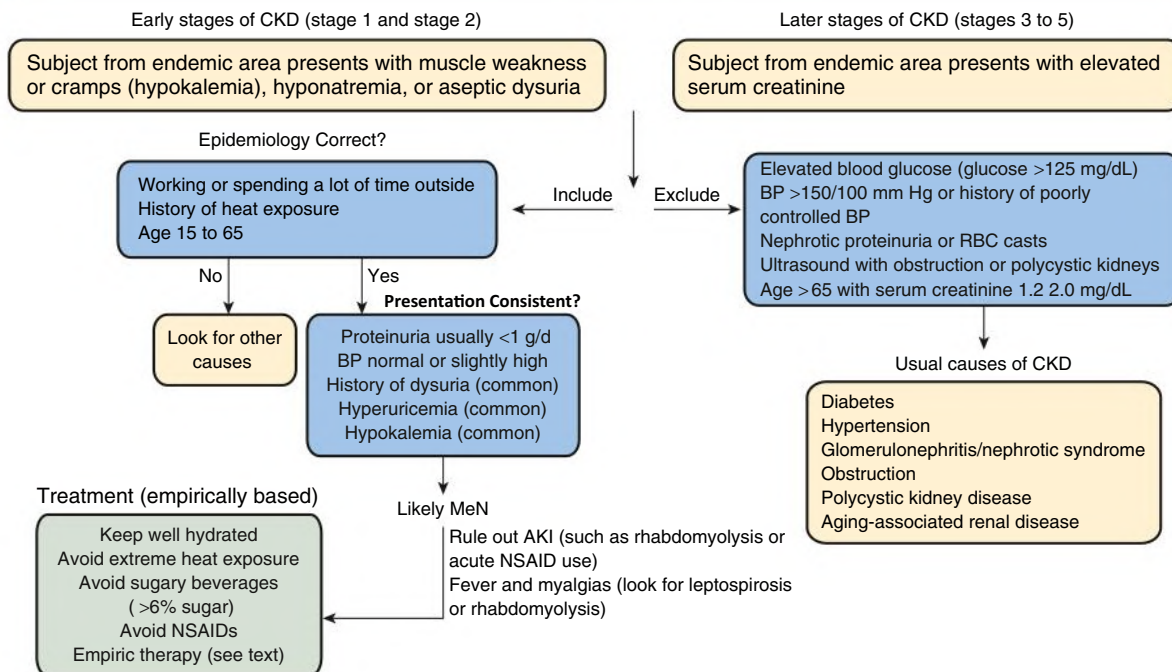


Fig. 66.4 Diagnosis and management of Mesoamerican nephropathy (MeN). *AKI*, Acute kidney injury; *BP*, blood pressure; *CKD*, chronic kidney disease; *NSAID*, nonsteroidal antiinflammatory drug; *RBC*, red blood cell.

because many workers believe the well water contains toxins, it seems likely that those who drink well water might be the same subjects who were limiting their water intake. Thus, these studies also raise the question whether heat stress and recurrent dehydration may be involved in this disease as well.¹⁰

Although a specific cause has not been identified, a publication from the Ministry of Health in Sri Lanka suggests that the incidence of CKD may be declining.²³

OTHER ENDEMIC NEPHROPATHIES

An epidemic of CKD has been reported in the Uddanam region of Andhra Pradesh, southern India, and may have been present since the early 1990s.²⁴ Uddanam nephropathy is observed primarily in men working in the agricultural fields where the primary crops are cashews, coconuts, and rice. Similar to the other epidemics, patients present with minimal proteinuria and elevated serum creatinine and have chronic tubulointerstitial disease on kidney biopsy.²⁵ Hyperuricemia is also common. Large-scale prospective efforts to investigate disease incidence and risk factors are underway.

Other similar epidemics are also emerging elsewhere. This includes an epidemic among agricultural workers within Tierra Blanca in the Veracruz region of Mexico, where heat stress also has been mentioned as a risk factor, and agricultural work was associated with low eGFR.^{26,27} Similarly, the Central Valley of California and Rio Grande Valley in Texas have among the highest incidence of end-stage kidney disease (ESKD) in the United States,²⁸ and these “hot spots” have also been linked to agricultural activity.²⁹ There are also early reports that similar epidemics may be manifesting in northeastern Thailand, Egypt, and Sudan. Indeed, to date there are no distinguishing features among the various epidemics, and many believe the disease may have similar etiologies worldwide.

POTENTIAL ROLE OF CLIMATE CHANGE AND GLOBAL WARMING

The similarity between the various epidemics has placed special emphasis on the role of heat stress and dehydration in CKD. The sites where these endemic nephropathies are reported are usually extremely hot and located between the equator and the Tropic of Cancer. Although mean temperatures have increased only 0.8°C in the last 50 years, the frequency of extreme heat events (exceeding the 99th percentile) has increased by nearly 75%.¹⁰ Remarkably, the sites where reports are emerging of epidemic CKD in India correlate with areas of increased extreme heat events such as heat waves (Fig. 66.5). This has led to the hypothesis that these epidemics may represent a type of heat stress nephropathy or that global warming may have a role in accelerating the epidemics. If true, the increase in frequency being observed is likely related to increasing world temperatures. It also remains possible that toxins may contribute to the various epidemics or may be acting in synergy with heat stress to cause CKD, as suggested by the presence of some Fanconi-like features in Mesoamerica and other locales.

To date, none of the proposed hypotheses has been able to explain the origin of these endemic nephropathies alone. Thus, two or more factors could be required.⁷ Many facts support a possible genetic or epigenetic role in the development of the disease: its localized endemic nature, frequent familial recurrence, Hispanic ethnicity selection in Mesoamerica, and its absence from other places where similarly severe environmental and occupational conditions are present.^{7,9} A successful research approach to these disorders will require a multidisciplinary approach, the establishment of systematic registries and reinforcing the local capacity for kidney biopsies, genetic and environmental studies, and controlled trials of current diagnosis and management protocols. Given the linkage with heat stress and climate, there is also a need to determine whether similar epidemics are occurring in other hot regions where similar conditions are present.

Sites of Endemic Nephropathy in India Correlate With Sites of Prolonged Heat Waves

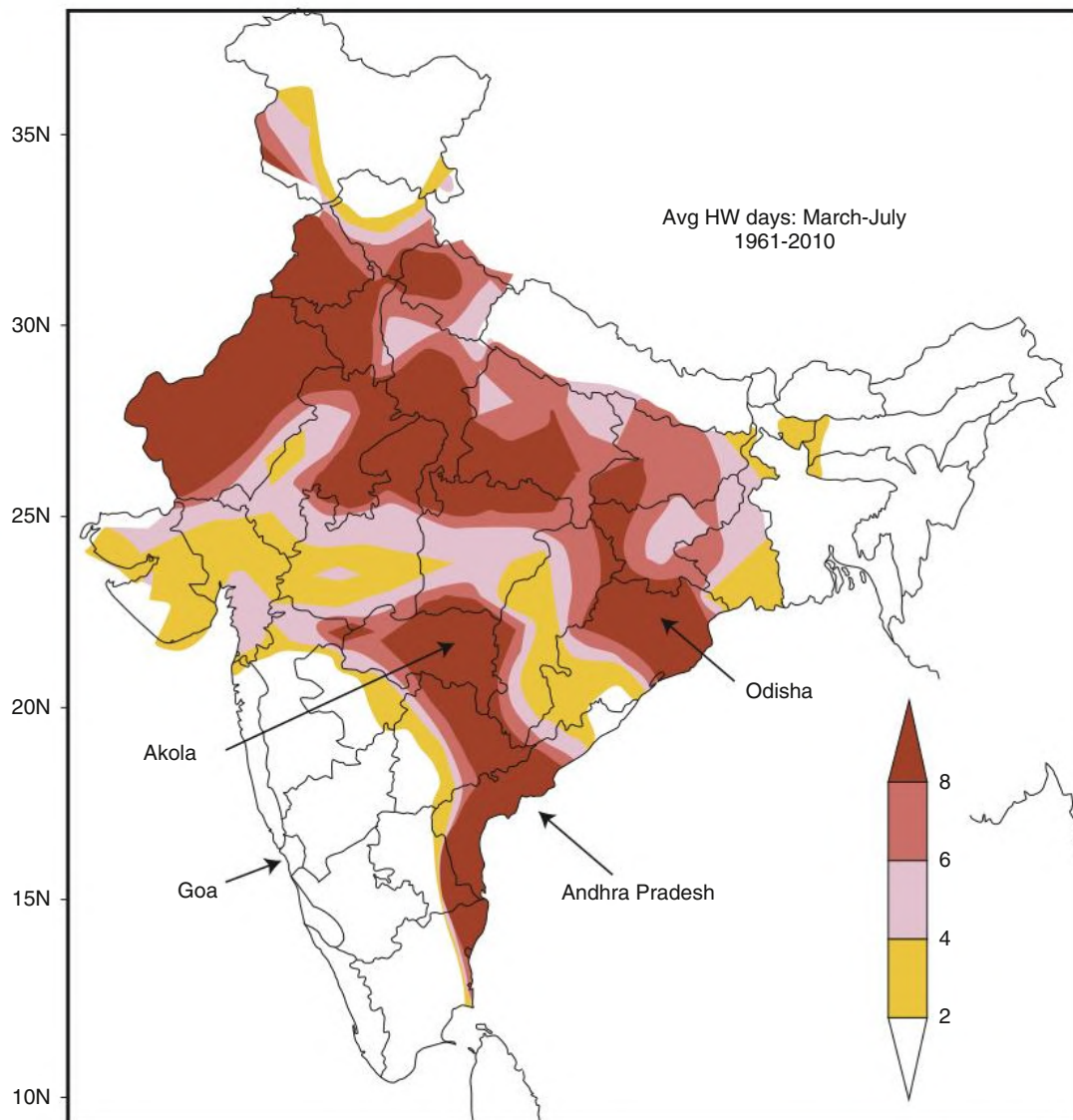


Fig. 66.5 Reported Sites in India With Increased Frequency of Chronic Kidney Disease (CKD) Correlate With Sites of Prolonged Heat Waves. Shown is the average number of heat wave days between March and July (hottest time of the year) in India, based on the number of heat wave days over the 50-year period. A heat wave is usually defined as sustained temperatures of greater than 40°C or an increase of 5 to 6°C over the normal maximal temperature of that region. Andhra Pradesh (including the Uddanam region) has had some of the longest heat waves, with one recorded at 35 days. Other suspected sites of CKD of unknown etiology include the Akola district of Maharashtra and the central Odisha region. *HW*, Heat wave. (Data from Pai DS, Nair S, Ramanathan AN. Long term climatology and trends of heat waves over India during the recent 50 years [1961–2010]. *MAUSAM*. 2013;64[4]:585–604; and Glaser J, Lemery J, Rajagopalan B, et al. Climate change and the emergent epidemic of chronic kidney disease from heat stress in rural communities: the case for heat stress nephropathy. *Clin J Am Soc Nephrol*. 2016;11[8]:1472–1483.)

SELF-ASSESSMENT QUESTIONS

1. Endemic nephropathies refer to:
 - A. regions where CKD of any cause is highly prevalent.
 - B. regions with a high prevalence of CKD that is unrelated to common causes such as hypertension and diabetes.
 - C. ethnic groups with high susceptibility to CKD.
 - D. ethnic groups with high prevalence of CKD.
 - E. regions with high prevalence of patients undergoing renal replacement therapy.
2. Which statement *best* describes Mesoamerican nephropathy?
 - A. A type of CKD of undetermined cause highly prevalent in the Pacific coast of Central America
 - B. A type of CKD caused by exposure to aristolochic acid
 - C. A type of CKD related to exposure to cadmium
 - D. A type of CKD often seen in Mesoamerican Maya ethnicity patients
 - E. A type of CKD that locals call *itai-itai*
3. Which of the following are similarities between Sri Lankan endemic nephropathy and Mesoamerican endemic nephropathy?
 - A. Highly symptomatic, predominantly glomerular damage with proteinuria, young patients from hot regions
 - B. Mostly asymptomatic, predominantly tubulointerstitial damage with low proteinuria, young workers from hot regions
 - C. Fanconi-like syndrome with high mortality rates and exposure to lead
 - D. Fanconi-like syndrome with high mortality rates and exposure to heat stress
 - E. Highly symptomatic, predominantly tubulointerstitial damage with proteinuria, elderly patients
4. The idea of a potential role of climate change and global warming is suspected in emerging endemic nephropathies because of which of the following?
 - A. Most emerging endemic regions correlate with areas of increased extreme heat.
 - B. Chronic kidney disease is much more common in workers in warmer areas despite similar working conditions.
 - C. Patients are exposed to extreme heat stress and dehydration; dehydration may be fueled by the fear of water contamination.
 - D. Most emerging endemic regions are located between the equator and the Tropic of Cancer.
 - E. All the above.

REFERENCES

- Garcia-Trabanino R, Aguilar R, Silva CR, et al. End-stage renal disease among patients in a referral hospital in El Salvador. *Pan Am J Public Health*. 2002;12(3):202–206.
- Wesseling C, van Wendel de Joode B, Crowe J, et al. Mesoamerican nephropathy: geographical distribution and time trends of chronic kidney disease mortality between 1970 and 2012 in Costa Rica. *Occup Environ Med*. 2015;72(10):714–721.
- Raines N, Gonzalez M, Wyatt C, et al. Risk factors for reduced glomerular filtration rate in a Nicaraguan community affected by Mesoamerican nephropathy. *MEDICC Rev*. 2014;16(2):16–22.
- Valcke M, Levasseur ME, Soares da Silva A, et al. Pesticide exposures and chronic kidney disease of unknown etiology: an epidemiologic review. *Environ Health*. 2017;16(1):49.
- Diyabalanage S, Fonseka S, Dasanayake DMSNB, et al. Environmental exposures of trace elements assessed using keratinized matrices from patients with chronic kidney diseases of uncertain etiology (CKDu) in Sri Lanka. *J Trace Elem Med Biol*. 2017;39:62–70.
- Schaeffer JW, Adgate JL, Reynolds SJ, et al. A pilot study to assess inhalation exposures among sugarcane workers in Guatemala: implications for chronic kidney disease of unknown origin. *Int J Environ Res Public Health*. 2020;17(16):5708.
- Sanchez Polo V, Garcia-Trabanino R, Rodriguez G, et al. Mesoamerican nephropathy (MeN): what we know so far. *Int J Nephrol Renovasc Dis*. 2020;13:261–272.
- Fischer RSB, Unrine JM, Vangala C, Sanderson WT, Mandayam S, Murray KO. Evidence of nickel and other trace elements and their relationship to clinical findings in acute mesoamerican nephropathy: a case-control analysis. *PLoS One*. 2020;15(11):e0240988.
- González-Quiroz M, Pearce N, Caplin B, Nitsch D. What do epidemiological studies tell us about chronic kidney disease of undetermined cause in Meso-America? A systematic review and meta-analysis. *Clin Kidney J*. 2018;11(4):496–506.
- Glaser J, Lemery J, Rajagopalan B, et al. Climate change and the emergent epidemic of chronic kidney disease from heat stress in rural communities: the case for heat stress nephropathy. *Clin J Am Soc Nephrol*. 2016;11(8):1472–1483.
- Wijkström J, Jayasumana C, Dassanayake R, et al. Morphological and clinical findings in Sri Lankan patients with chronic kidney disease of unknown cause (CKDu): similarities and differences with Mesoamerican Nephropathy. *PLoS One*. 2018;13(3):e0193056.
- Fischer RSB, Vangala C, Truong L, et al. Early detection of acute tubulointerstitial nephritis in the genesis of Mesoamerican nephropathy. *Kidney Int*. 2018;93(3):681–690.
- Herrera R, Orantes CM, Almaguer M, et al. Clinical characteristics of chronic kidney disease of nontraditional causes in Salvadoran farming communities. *MEDICC Rev*. 2014;16(2):39–48.
- Kupferman J, Amador JJ, Lynch KE, et al. Characterization of Mesoamerican nephropathy in a kidney failure hotspot in Nicaragua. *Am J Kidney Dis*. 2016;68(5):716–725.
- Garcia-Trabanino R, Hernandez C, Rosa A, et al. Incidence, mortality, and prevalence of end-stage chronic renal disease in the Bajo Lempa region of El Salvador: a ten-year community registry. *Nefrologia*. 2016;36(5):517–522.
- Wimalawansa SJ. Escalating chronic kidney diseases of multifactorial origin (CKD-MFO) in Sri Lanka: causes, solutions, and recommendations-update and responses. *Environ Health Prev Med*. 2015;20(2):152–157.
- Anand S, Montez-Rath ME, Adasooriya D, et al. Prospective biopsy-based study of CKD of unknown etiology in Sri Lanka. *Clin J Am Soc Nephrol*. 2019;14(2):224–232.
- Wijetunge S, Ratnatunga NV, Abeysekera TD, et al. Endemic chronic kidney disease of unknown etiology in Sri Lanka: correlation of pathology with clinical stages. *Indian J Nephrol*. 2015;25(5):274–280.
- Nanayakkara N, Wazil AWM, Gunerathne L, et al. Tackling the fallout from chronic kidney disease of unknown etiology: why we need to focus on providing peritoneal dialysis in rural, low-resource settings. *Kidney Int Rep*. 2016;2(1):1–4.
- Rango T, Jeuland M, Manthirithilake H, McCornick P. Nephrotoxic contaminants in drinking water and urine, and chronic kidney disease in rural Sri Lanka. *Sci Total Environ*. 2015;574–585. 518–519.
- Nanayakkara S, Senevirathna STMLD, Parahitiyawa NB, et al. Whole-exome sequencing reveals genetic variants associated with chronic kidney disease characterized by tubulointerstitial damages in North Central Region, Sri Lanka. *Environ Health Prev Med*. 2015;20(5):354–359.
- Siriwardhana EA, Perera PA, Sivakanesan R, et al. Dehydration and malaria augment the risk of developing chronic kidney disease in Sri Lanka. *Indian J Nephrol*. 2015;25(3):146–151.
- Ranasinghe AV, Kumara GWGP, Karunaratna RH, et al. The incidence, prevalence and trends of chronic kidney disease and chronic kidney disease of uncertain aetiology (CKDu) in the north central province of Sri Lanka: an analysis of 30,566 patients. *BMC Nephrol*. 2019;20(1):338.
- Reddy DV, Gunasekar A. Chronic kidney disease in two coastal districts of Andhra Pradesh, India: role of drinking water. *Environ Geochem Health*. 2013;35(4):439–454.
- Ganguli A. Uddanam nephropathy/regional nephropathy in India: preliminary findings and a plea for further research. *Am J Kidney Dis*. 2016;68(3):344–348.
- Aguilar-Ramirez D, Raña-Custodio A, Villa A, et al. Decreased kidney function and agricultural work: a cross-sectional study in middle-aged adults from Tierra Blanca, Mexico. *Nephrol Dial Transplant*. 2020: gfaa041.
- Aguilar DJ, Madero M. Other potential CKD hotspots in the world: the cases of Mexico and the United States. *Semin Nephrol*. 2019;39(3):300–307.
- Bragg-Gresham J, Morgenstern H, Shahinian V, et al. An analysis of hot spots of ESRD in the United States: potential presence of CKD of unknown origin in the USA? *Clin Nephrol*. 2020;93(1):113–119.
- Anand S, Staniec A, Montez-Rath M, et al. Using GIS mapping to track hot spots of kidney disease in California. *N Engl J Med*. 2020;382(23):2265–2267.